

AWS Cloud Fundamentals

SkillNexus — Assignment Report

Assignments Covered

1. Create and Configure AWS Account
2. Explore Core AWS Services

This report documents the completion of the AWS account configuration tasks and provides a conceptual summary of the core AWS services requested in the assignment. Assignment 2 is limited to identification and description of the services; no separate resource deployment is required for that section.

Assignment 1 — Create and Configure AWS Account

Objective

To configure an AWS Free Tier account, create an IAM user, assign appropriate permissions through an IAM group/policy, create an IAM role, and explore the AWS Management Console.

1. AWS Free Tier Account

An AWS Free Tier account provides access to eligible AWS services and usage limits intended for learning, experimentation, and development. The AWS Management Console was accessed successfully for the account.

2. IAM User Creation

An IAM user named **Aditya-SkillNexus** was created. The user was placed in the **SkillNexus-Developers** group. The permissions view shows the AWS managed policy **AmazonEC2ReadOnlyAccess** attached through that group.

3. IAM Role Creation

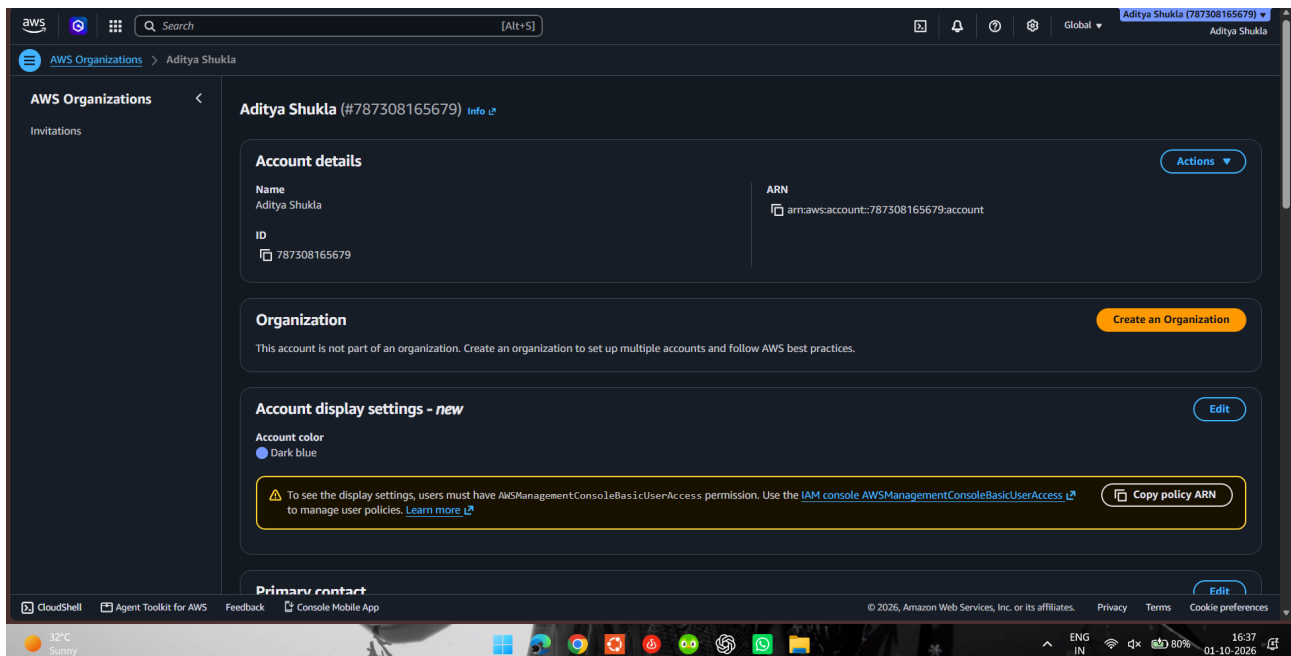
An IAM role named **SkillNexus-EC2-Role** was created using the EC2 service use case. The role is intended to allow an EC2 instance to obtain permissions to interact with AWS services through role-based access rather than embedding long-term credentials on the instance.

4. AWS Management Console Exploration

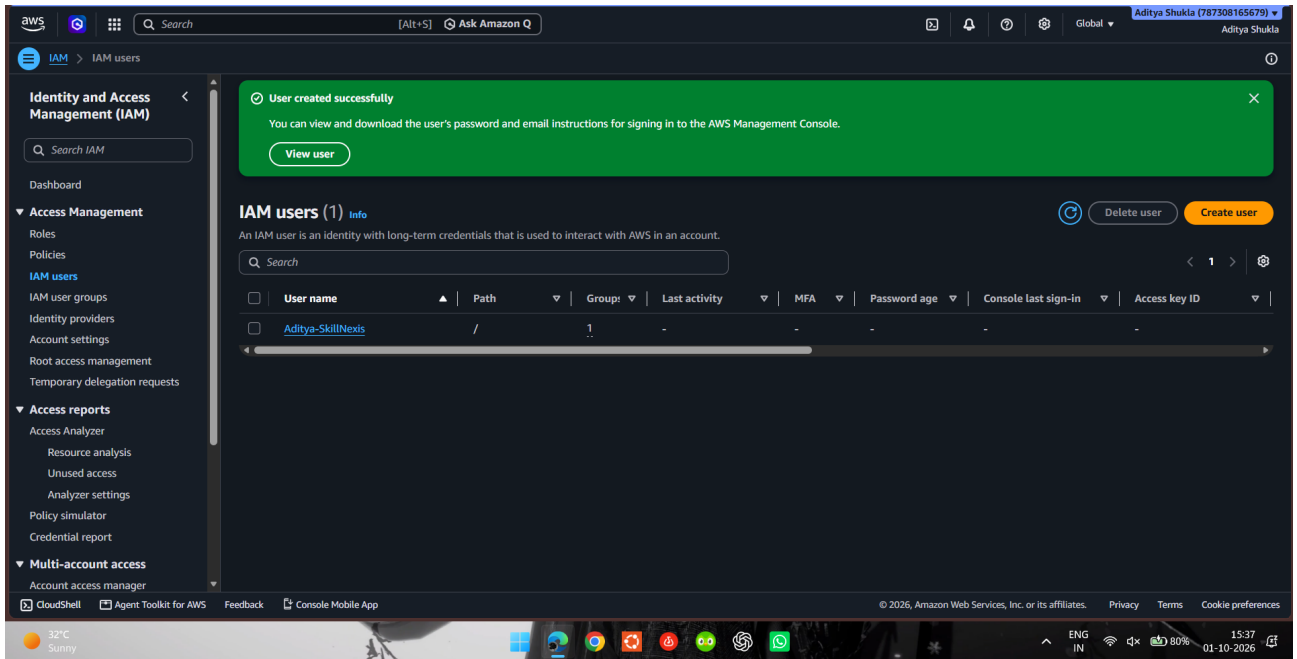
The AWS Management Console was explored through IAM and EC2. The screenshots document the IAM users, IAM role, and an EC2 instance running in the AWS console.

Evidence / Screenshots

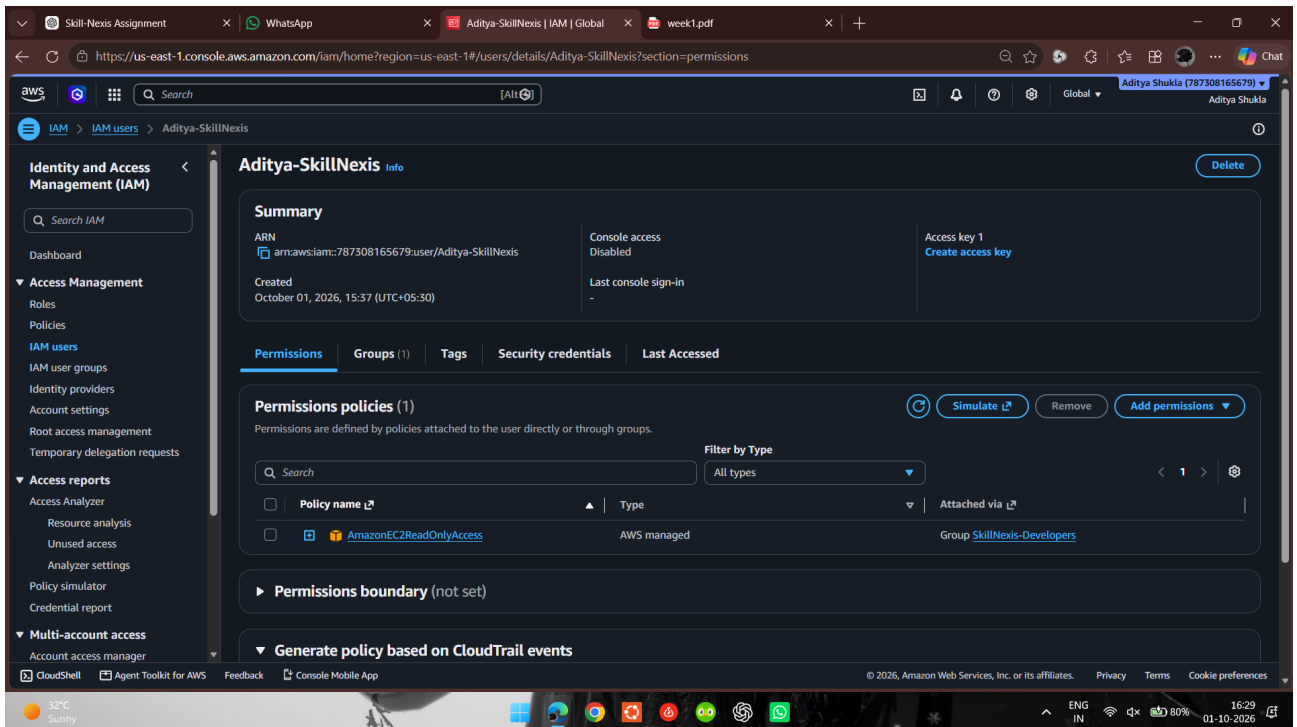
AWS Free Tier Account



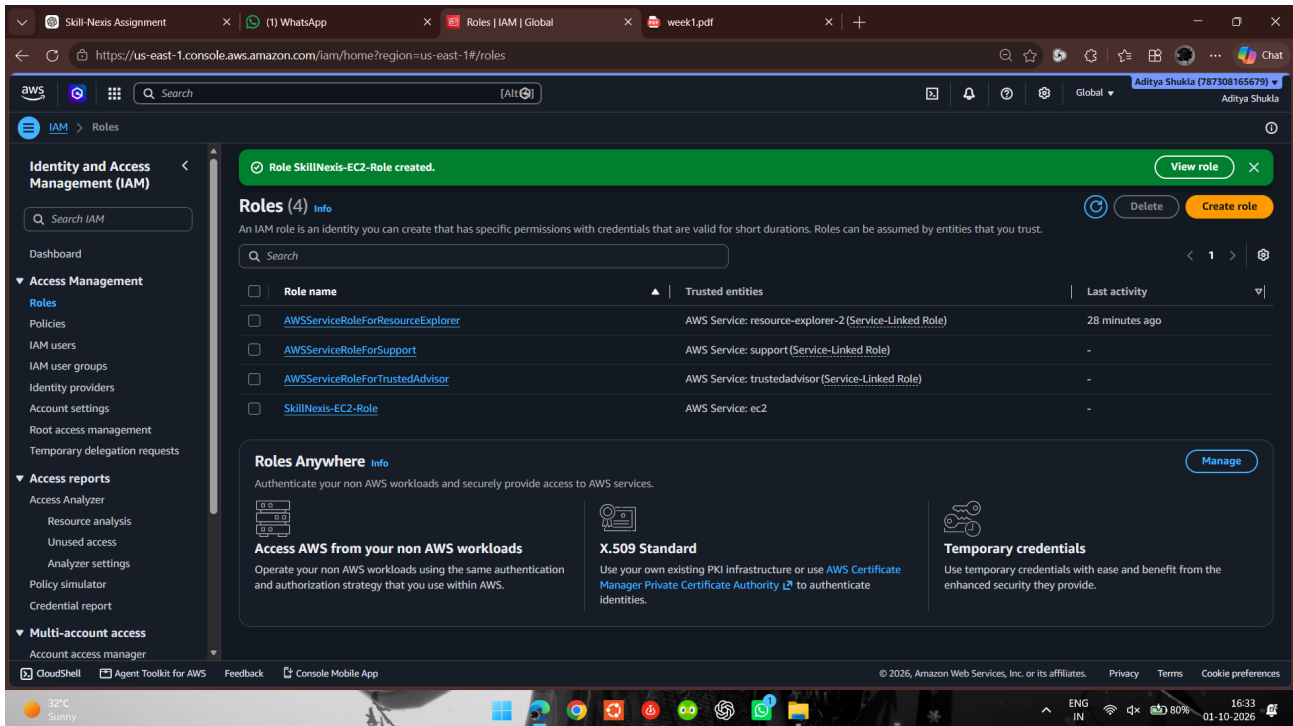
IAM User Created



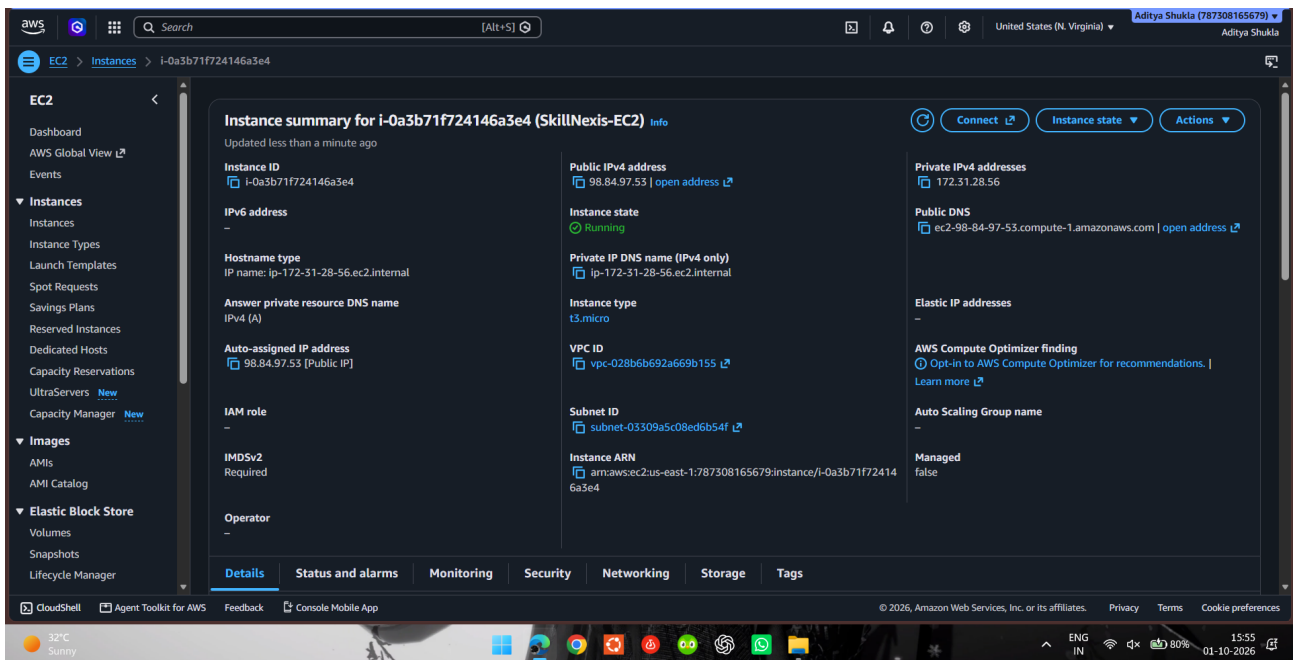
IAM User Permissions / Details



IAM Role Created



AWS Management Console / EC2



Assignment 2 — Explore Core AWS Services

Objective

Identify the four core AWS services specified in the assignment and summarize their primary functions and typical use cases.

Service	Primary Function	Typical Use
Amazon EC2	Elastic Compute Cloud provides resizable virtual servers (instances) in the AWS cloud. Users can choose operating systems, instance types, storage, and networking settings.	Running applications, backend servers, APIs, development environments, and other workloads that require compute capacity.
Amazon S3	Simple Storage Service is an object storage service designed to store and retrieve files and other objects using buckets.	Storing images, documents, backups, static website assets, logs, datasets, and application files.
Amazon RDS	Relational Database Service is a managed database service that simplifies setup, operation, backups, scaling, and maintenance of supported relational database engines.	Hosting application databases while reducing the amount of manual database administration required.
AWS Lambda	Lambda is a serverless compute service that runs code in response to events without requiring the user to manage servers.	Event-driven processing, APIs, automation, scheduled jobs, file processing, and lightweight backend logic.

Service Relationship — Quick View

EC2 → Compute: runs the application/server.

S3 → Storage: stores objects/files.

RDS → Database: provides managed relational databases.

Lambda → Serverless Compute: runs code on demand in response to events.

Conclusion

The required AWS account configuration tasks were completed and documented with screenshots. The core AWS services identified in Assignment 2 were reviewed at a conceptual level, with their primary functions and common uses summarized above.